# Zichen Gui

*Curriculum Vitae*

## Position Held

2024–present **Assistant Professor**, *University of Georgia.*

## Postdoctoral Experience

2022–2024 **Postdoctoral Fellow**, *ETH Zürich.*
Host - Prof. Kenneth G. Paterson.

## Education

2017–2022 **Ph.D. in Computer Science**, *University of Bristol.*
Advisors - Prof. Oliver Johnson and Prof. Bogdan Warinschi.

2013–2017 **M.Eng in Mathematics and Computer Science**, *University of Bristol.*
Graduated with First Class Honours.

## Professional Experience

2020–2021 **Research Intern**, *ETH Zürich.*
Host - Prof. Kenneth G. Paterson.

## Publications

2026 **Page-efficient Encrypted Multi-Maps: New Techniques for Optimal Search Bandwidth.**
Asiacrypt 2026.
Francesca Falzon, Zichen Gui, Michael Reichle.

2026 **DDR-SSE: Duplicated Retrieval of Documents for System-wide Secure Searchable Symmetric Encryption.**
USENIX Security Symposium 2026.
Zichen Gui, Simon-Philipp Merz, Kenneth G. Paterson, Sikhar Patranabis.

2026 **Encrypted Fringe Projection Profilometry.**
Optics Letters.
Yang Cheng, Zichen Gui, Le Guan, Ran Jin, Beiwen Li.

2024 **Understanding Leakage in Searchable Encryption: a Quantitative Approach.**
Proc. Priv. Enhancing Technol. (PoPETS) 2024.
Alexandra Boldyreva, Zichen Gui, Bogdan Warinschi.

2024 **SWiSSSE: System-Wide Security for Searchable Symmetric Encryption.**
Proc. Priv. Enhancing Technol. (PoPETS) 2024.
Zichen Gui, Kenneth G. Paterson, Sikhar Patranabis, Bogdan Warinschi.

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2023 **Security Analysis of MongoDB Queryable Encryption.**
USENIX Security 2023.
Zichen Gui, Kenneth G. Paterson, Tianxin Tang.

2023 **Rethinking Searchable Symmetric Encryption.**
IEEE S&P 2023.
Zichen Gui, Kenneth G. Paterson, Sikhar Patranabis.

2019 **Encrypted Databases: New Volume Attacks against Range Queries.**
ACM CCS 2019.
Zichen Gui, Oliver Johnson, Bogdan Warinschi.

## Talks

2026 **DDR-SSE: Duplicated Retrieval of Documents for System-wide Secure Searchable Symmetric Encryption.**
USENIX Security Symposium 2026.

2024 **Understanding Leakage in Searchable Encryption: a Quantitative Approach.**
Proc. Priv. Enhancing Technol. (PoPETS) 2024.

2024 **SWiSSSE: System-Wide Security for Searchable Symmetric Encryption.**
Proc. Priv. Enhancing Technol. (PoPETS) 2024.

2023 **Rethinking Searchable Symmetric Encryption.**
IEEE S&P 2023, Encryption for Secure Search and other Algorithms 2023.

2019 **Encrypted Databases: New Volume Attacks against Range Queries.**
ACM CCS 2019.

## Professional Activities and Services

- **Conference Program Committee Memberships:** ACM CCS 2023-2026, PETS 2026-2027, ACM AsiaCCS 2026, USENIX Security 2026-2027, SPACE 2026.
- **External Reviewing - Conferences:** IACR Eurocrypt, IACR Asiacrypt, IEEE S&P.

## Teaching

- **Introduction to Cryptography:** 2025-2026.
- **Advanced Topics in Cryptography:** 2025-2026.

## Languages and Programming Languages

- Fluent in English. Native Chinese speaker.
- Experienced in C, C++, Java, JavaScript, Python and SQL.